Ad Copilot — Architecture & API Integration Doc

Status: Draft v1.0 **Owner:** Product, in partnership with Engineering **Related docs:** Case Study Narrative · PRD · User Stories · AI Workflow Design · Metrics & Attribution Spec · Release Plan (next)

This doc is written as the artifact a PM brings *into* the engineering design review — enough system detail to have a real conversation about tradeoffs, not enough to substitute for engineering's own design doc. Where a decision is genuinely engineering's call, it's flagged as an open question rather than dictated.

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1. System Overview

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                    ┌──────────────────────┐
                    │   Ad Platforms        │
                    │  (Meta Ads, Google    │
                    │   Ads APIs)           │
                    └──────────┬───────────┘
                               │ OAuth + REST
                               ▼
                    ┌──────────────────────┐
                    │  Ingestion Service    │
                    │  (scheduled hourly    │
                    │   pull, rate-limit    │
                    │   aware)              │
                    └──────────┬───────────┘
                               ▼
                    ┌──────────────────────┐
                    │  Performance Data     │
                    │  Store (time-series)  │
                    └──────────┬───────────┘
                               ▼
                    ┌──────────────────────┐
                    │  Drift Detection &    │
                    │  Recommendation Svc   │
                    └──────────┬───────────┘
                               │
             ┌─────────────────┼─────────────────┐
             ▼                                    ▼
   ┌──────────────────────┐          ┌──────────────────────┐
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│  Creative Generation  │            │  Confidence Scoring   │
│  Service (LLM calls,  │◄───────────┤  & Lane Assignment    │
│  tiered routing)      │            │  Service              │
└───────────┬───────────┘            └───────────┬───────────┘
            │                                    │
            ▼                                    ▼
┌──────────────────────┐            ┌──────────────────────┐
│  Brand-Safety Filter  │            │  Review Queue         │
│  (deterministic)      │───────────►│  (UI + API)           │
└───────────────────────┘            └───────────┬───────────┘
                                                 ▼
                                      ┌──────────────────────┐
                                      │  Action Executor      │
                                      │  (writes back to Ad   │
                                      │   Platform APIs)      │
                                      └──────────┬────────────┘
                                                 ▼
                                      ┌──────────────────────┐
                                      │  Audit Log & Rollback│
                                      └──────────────────────┘
```

2. Core Components

Component	Responsibility
Ingestion Service	Pulls campaign/performance data from Meta and Google Ads APIs on an hourly schedule; handles OAuth token refresh, rate-limit backoff, and pagination
Performance Data Store	Time-series storage of spend, conversions, ROAS/CPA/CPC/CTR per campaign/ad set, retained at sufficient granularity for drift-window calculations
Drift Detection & Recommendation Service	Compares live metrics against targets; generates recommendation objects with the triggering signal and proposed action
Creative Generation Service	Manages tiered LLM calls (first-pass + refinement per AI Workflow Design §2), including caching and per-account cap enforcement
Confidence Scoring & Lane Assignment Service	Computes the composite confidence score (AI Workflow Design §4) and assigns Auto-apply/Suggest/Escalate
Brand-Safety Filter	Deterministic, non-model check run on all generated copy before it

	can enter the review queue
Review Queue (UI + API)	Marketer-facing surface for approve/edit/reject; also the system-of-record for human feedback used in the calibration loop
Action Executor	Writes approved/auto-applied changes back to the ad platform APIs; the only component with write access to live campaigns
Audit Log & Rollback	Immutable log of every AI action across all lanes; supports the 5-minute rollback window for Auto-apply actions (NFR-2)

3. Key API Integrations

3.1 Meta Marketing API

- **Auth:** OAuth 2.0, long-lived system-user tokens where available to reduce re-auth friction for the marketer; token refresh handled by the Ingestion Service, not the client.
- **Reads:** Campaign/Ad Set/Ad structure, Insights API for performance metrics (spend, conversions, ROAS proxy via purchase value, CTR, CPC).
- **Writes:** Budget/bid updates via the Ad Set update endpoint (Action Executor only); new ad copy creation via the Ad Creative endpoint.
- **Known constraint:** Insights API data can be provisional for up to ~72 hours (ties directly to Metrics & Attribution Spec §4) — the ingestion layer tags data with a “final vs. provisional” flag pulled from the API response where available, and this flag propagates into recommendation reasoning.
- **Rate limits:** tiered by app usage; Ingestion Service implements exponential backoff and a request queue rather than naive retry, and stales-out gracefully into the “data freshness” UI indicator (NFR-1) rather than blocking the whole account’s sync.

3.2 Google Ads API

- **Auth:** OAuth 2.0 + Google Ads API developer token; note Google’s token/developer-token approval process has its own lead time — flagged early to engineering as a launch-timeline dependency, not something to discover during Phase 2 build.
- **Reads:** GAQL (Google Ads Query Language) queries against campaign/ad group/ad performance tables.
- **Writes:** Budget/bid mutations via the Google Ads mutate API; responsible_growth_notice: Google’s API enforces stricter change-rate limits on mutates than Meta’s, which directly informs how aggressively Auto-apply can act on

Google campaigns (a tighter blast-radius default may be warranted here — flagged as an open question, §6).

- **Known constraint:** attribution model differences from Meta (data-driven attribution vs. Meta's click/view windows) reinforce the platform-native, non-unified attribution decision in the Metrics & Attribution Spec §3.

3.3 LLM Provider API

- **Calls:** structured-output requests (JSON schema-constrained) for both creative generation and recommendation-reasoning tasks, per AI Workflow Design §3.
- **Tiering:** first-pass generation and refinement calls route to different model endpoints/tiers (AI Workflow Design §2); routing logic lives in the Creative Generation Service, not scattered across callers.
- **Cost/latency tracking:** every call is logged with token count, model tier, and prompt-template version (NFR-4), feeding both the cost-cap enforcement (US-12) and the calibration/versioning work in AI Workflow Design §7.
- **Failure handling:** circuit breaker pattern — if error rate/latency crosses a threshold, the service degrades to "generation delayed" rather than retrying aggressively into a provider incident (AI Workflow Design §6).

4. Data Model Notes (illustrative, not a full schema)

- **Account** — platform connections, trust tier, configured thresholds (per US-11), plan tier/cost cap.
- **Campaign/AdSet** — target metric type and value, current lane-eligibility state.
- **Recommendation** — triggering signal, proposed action, confidence score + component breakdown, lane, status (pending/approved/edited/rejected/auto-applied/rolled back), model/prompt version.
- **GeneratedCreative** — variant text, confidence score, brand-safety check result, cache key (feed-content hash + prompt-template version).
- **AuditEvent** — immutable append-only record of every state change to a Recommendation or GeneratedCreative, timestamped, tied to the acting entity (system vs. specific user).

5. Security & Compliance Considerations

- Ad platform OAuth tokens stored encrypted at rest; scoped to minimum required permissions (read insights + write budget/creative, not full account admin).
- Action Executor is the only component with write credentials to ad platforms — this isolation is deliberate so that a bug in generation or scoring logic cannot directly mutate live campaigns without passing through lane assignment and (where required) human approval.
- Audit log is append-only and not user-editable, since it's the record of truth for both customer trust and any future dispute about "why did the AI do this."

6. Open Questions for Engineering

- Should Google Ads campaigns get a stricter default blast-radius cap than Meta, given the platform's tighter mutate rate limits (§3.2)? Product's instinct is yes, but the exact number should come from a technical spike, not a product guess.
- What's the right storage/retention window for the Performance Data Store to support drift-window calculations without unbounded time-series growth per account?
- Does the LLM provider's structured-output mode give hard schema guarantees, or does the Creative Generation Service need its own validation/retry layer on malformed responses? (AI Workflow Design assumes validation happens somewhere — this doc doesn't yet commit to where.)
- Google Ads developer token approval lead time — needs to be confirmed against the Phase 2 timeline in the Release Plan before that phase is scheduled with confidence.